

Mode Collapse as an Optimal Transport Problem in Variational Inference

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Abstract

This note explores the reformulation of variational inference (VI) as an optimal transport (OT) problem, providing a geometric perspective on mode collapse. By interpreting VI as the task of finding an efficient transport plan between the variational distribution and the target distribution, we define mode collapse as the degeneracy of such transport plans. We introduce the concept of a mode-covering transport plan and characterize mode collapse through its failure conditions. This framework unifies different collapse mechanisms and suggests new theoretical and algorithmic directions.

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1 Introduction

Variational inference approximates a target distribution p by minimizing a divergence (typically the reverse Kullback-Leibler divergence) over a tractable family $\mathcal{Q}$. A common pathology is *mode collapse*, where the variational distribution fails to capture all significant regions (modes) of p , even when the family is expressive enough. We reinterpret VI as an optimal transport problem, where the goal is to transport probability mass from the variational distribution to the target distribution efficiently. This perspective provides a geometric understanding of mode collapse and leads to precise definitions and diagnostic tools.

2 Variational Inference as Optimal Transport

Let p be the target distribution and $\mathcal{Q}$ a variational family. We consider a cost function $c : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}^+$ and a regularization term $R(\pi)$ with strength λ . The OT formulation of VI is:

$$q^* = \arg \min_{q \in \mathcal{Q}} \inf_{\pi \in \Pi(p, q)} \left\{ \int c(x, y) d\pi(x, y) + \lambda R(\pi) \right\}$$

where $\Pi(p, q)$ is the set of joint distributions with marginals p and q . Regularization (e.g., entropic regularization) ensures computational tractability and encourages diverse transport routes.

2.1 Special Case: Standard VI

When $c(x, y) = -\log p(x)$ and $R(\pi) = D_{\text{KL}}(\pi \| p \otimes q)$ with $\lambda = 1$, we recover standard VI:

$$q^* = \arg \min_{q \in \mathcal{Q}} D_{\text{KL}}(q \| p)$$

2.2 Intuition

VI becomes the problem of finding the cheapest way to transport probability mass from q to p :

- **Transport cost:** effort to reshape q into p
- **Variational family:** constraints on available “transport vehicles”
- **Mode collapse:** some regions of p are too expensive to supply
- **Regularization:** encourages a broader class of transport plans

3 Mathematical Definitions of Modes

To formalize mode collapse, we first define modes rigorously.

3.1 Topological Definition

Definition 1 (Mode). A point $x^* \in \mathcal{X}$ is a mode of p if:

1. $p(x^*) > 0$
2. There exists an open neighborhood $U \ni x^*$ such that $p(x^*) \geq p(x)$ for all $x \in U$
3. x^* is isolated: there exists $\epsilon > 0$ such that the ball $B_\epsilon(x^*)$ contains no other point satisfying conditions 1 and 2.

3.2 Practical Definition via -Modes

For analytical and computational purposes, we define -modes.

Definition 2 (-Mode). An -mode is a connected component $M \subset \mathcal{X}$ such that:

1. **Significant mass:** $\mathbb{P}_{x \sim p}(x \in M) \geq \delta$
2. **Peak condition:** There exists $x^* \in M$ such that $p(x^*) \geq (1 + \alpha) p(x)$ for all x on the boundary of M
3. **Separation:** For any two -modes M and M' , the distance between their peaks is at least Δ

where $\delta, \alpha, \Delta > 0$ are tolerance parameters.

3.3 Alternative: High-Probability Regions

A mode can also be defined as a connected component of a high-probability region

$$\{x \in \mathcal{X} : p(x) \geq \tau\}$$

for a threshold τ . The choice of definition depends on the context, but for Gaussian mixtures, these definitions effectively capture the intuitive notion of modes.

4 Mode-Covering Transport Plan

Definition 3 (Mode-Covering Transport Plan). A transport plan $\pi \in \Pi(p, q)$ is mode-covering if for every -mode M_i of p , there exists a region $R_i \subset \mathcal{X}$ such that:

1. **Mass preservation:** $\pi(M_i \times R_i) \geq (1 - \gamma) p(M_i)$
(sufficient mass from M_i is delivered to R_i)
2. **Concentration:** $q(R_i) \geq (1 - \gamma) p(M_i)$
(enough variational mass resides in R_i to receive the delivery)
3. **Fidelity:** $\sup_{x \in M_i, y \in R_i} c(x, y) \leq \epsilon$
(the regions M_i and R_i are geometrically close)

The parameters $\gamma, \epsilon > 0$ control the strictness of the conditions.

5 Mode Collapse Definition

Definition 4 (Mode Collapse). *Mode collapse occurs when no mode-covering transport plan exists between p and any $q \in \mathcal{Q}$.*

Equivalently, for every transport plan π , at least one mode M_i fails one of the three conditions:

- **Geometric failure:** $\sup_{x \in M_i, y \in R_i} c(x, y) > \epsilon$ (no variational component is sufficiently close to the target mode)
- **Mass deficiency:** $q(R_i) < (1 - \gamma) p(M_i)$ (insufficient variational mass near the target mode)
- **Routing failure:** $\pi(M_i \times R_i) < (1 - \gamma) p(M_i)$ (the transport plan does not allocate enough mass to cover the mode)

This definition unifies different collapse phenomena (mean alignment, vanishing weights, insufficient capacity) as degeneracies in the optimal transport plan.

6 Theoretical Goals and Conjectures

We outline several theoretical results to pursue.

6.1 Connection to KL Divergence

Conjecture 1. *Existence of a mode-covering transport plan implies a bounded KL divergence. Formally, if there exists a mode-covering transport plan between p and q , then*

$$D_{KL}(q||p) \leq f(\gamma, \epsilon)$$

for some function f that increases with γ and ϵ . This would justify the mode-covering condition as a sufficient criterion for good approximation.

6.2 Critical Separation

Conjecture 2. *For a family of target distributions parameterized by mode separation A , there exists a critical separation A_c such that:*

- *For $A < A_c$, a mode-covering transport plan exists (no collapse)*
- *For $A > A_c$, no mode-covering plan exists (collapse inevitable)*

This would formalize the empirical observation that mode collapse becomes harder to avoid as modes become more separated.

6.3 Inevitability of Collapse

Goal: Characterize scenarios where mode collapse is measure-theoretically inevitable, i.e., the set $\{q \in \mathcal{Q} : \text{no mode collapse}\}$ has measure zero. This would identify fundamental limitations of certain variational families.

6.4 Algorithmic Implications

Goal: Develop algorithms that explicitly search for mode-covering transport plans. Since a mode-covering plan guarantees good approximation, such algorithms would naturally avoid collapse.

7 Conclusion

Reformulating variational inference as an optimal transport problem provides a geometric lens through which mode collapse can be precisely defined and analyzed. The concept of a mode-covering transport plan captures the essence of successful approximation and leads to a unified characterization of collapse. This perspective suggests new theoretical questions and algorithmic strategies, potentially advancing the understanding and mitigation of mode collapse in complex inference tasks.

References

- “A theoretical perspective on mode collapse in variational inference” (Soletskyi, Gabriel, Loureiro)
- Optimal transport: Villani (2009), Peyré & Cuturi (2019)
- Variational inference: Blei et al. (2017)